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CENTRAL ELECTRICITY AUTHORITY

Session : Technical Paper Presentation

AI/ML & RPA LED DIGITAL TRANSFORMATION IN TL DESIGN

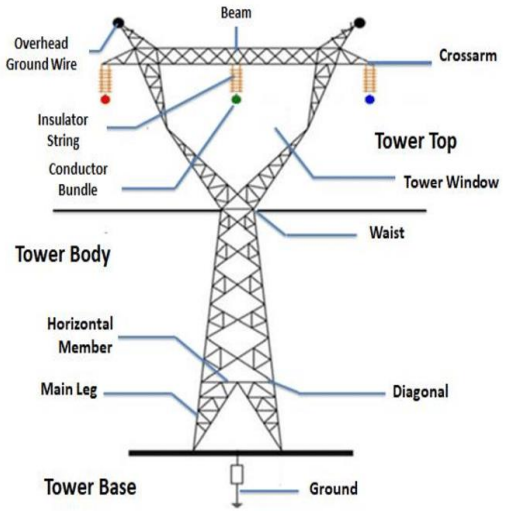
Presented By

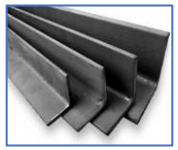
AVNISH GUPTA, MANAGER, KEC INTERNATIONAL LIMITED

INTRODUCTION

Transmission towers play a fundamental role in electric power distribution – powering homes, businesses and entire communities.

Various components involved & Complexities





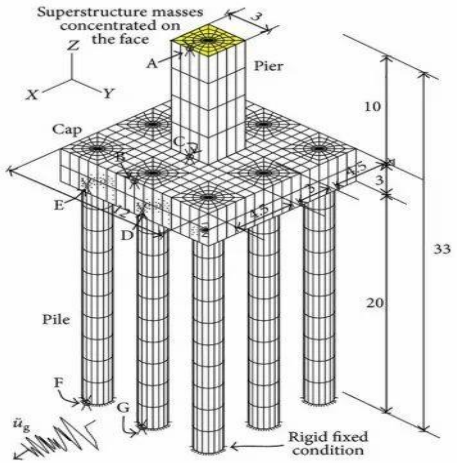
Tower Steel Structures

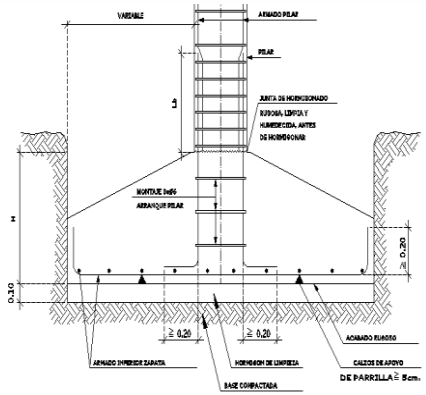
400+ Load cases per tower type

300+ towers of ~5-10 tower types

500+ patterns

250+ section sizes





Foundations

20+ Soil classes across codes

5+ Codal provisions

30+ shapes & design combinations

CONTEXT

Traditional TL process

A Tender / Project initiation

B Initial concept and design

C Tower drawings

D BOM generation

E Proto manufacturing and testing

Limitations of Traditional TL design process

Manual Geometry Modelling	Designers manually model tower configurations in PLS-Tower consuming time
Spreadsheet-Driven Calculations	Structural load computations, wind/temperature cases relying on spreadsheets that require experience and have risk of errors
Limited optimization	Manually iterate through various configurations & dimensions resulting in high cycle time and unoptimized cost
CAD drafting effort	Repetitive manual tasks to finalize drawings are time intensive

AI/ML & RPA LED DIGITAL TRANSFORMATION IN TL DESIGN (1/3)

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Business Needs

- Reduce project timelines
- Optimize tower weights (steel) and foundation volumes (concrete & steel)
- Improve standardization
- Enhance designer productivity amid talent shortages

Transformed TL design process

Digital Interventions

A Tender / Project initiation

ML & LLM Led – Quantity Estimators for Transmission Lines tenders

B Initial concept and design

Python based Automated Calculator

RPA based 3D Transmission Tower Modelling, Simulation & Optimization Suite

ML Led Foundation Design Optimization

C Tower drawings

Algorithm based single click refinement of Tower Drawing

Algorithm based single click generation of Foundation Drawing

D BOM generation

Algorithm-based single click BOM generation

E Proto manufacturing and testing

Document Management System

AI/ML & RPA LED DIGITAL TRANSFORMATION IN TL DESIGN (2/3)

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RPA based 3D Transmission Tower Modelling, Simulation & Optimization Suite

Driving inputs

Project specifications data

- Wire data
- Accessory data
- Climatic conditions
- Metal clearances
- Material Standard
- Bolt Standard
- Loading Standard
- Design Standard

Data Re-use from Standards Library

International Material Standards

Steel Grade

Bolt Grade

International Design & Loading Standards

Pattern (Geometry) Library

Customization

Member grouping

Integrated Design Tool

Metal Clearances

3D Clearance checks

Loads

Geometry generator

Iterative FE analysis

Design optimizer

Design parameters, geometries, patterns

Mathematical model for Cost optimization

Multiple configurations

Vertical

Horizontal

Varied dimensions

Base width

Hamper widths

Varied Panels

Cloud based library of 500+ patterns based on all possible standards

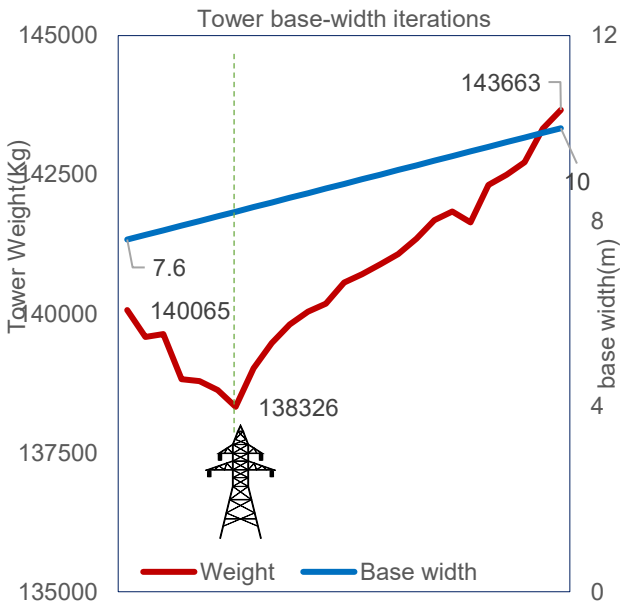
Technology

Robotic process automation tool designed on Unified Windows Platform (UWP) to automate design processes :

- 3D Geometry generation
- 3D model Analysis
- Load case creation
- 3D Clearance checks

Output

Digital Solution evaluates multiple tower configurations with different base widths and patterns, plots their weights and foundation forces, and helps engineers identify the lowest feasible tower weight.

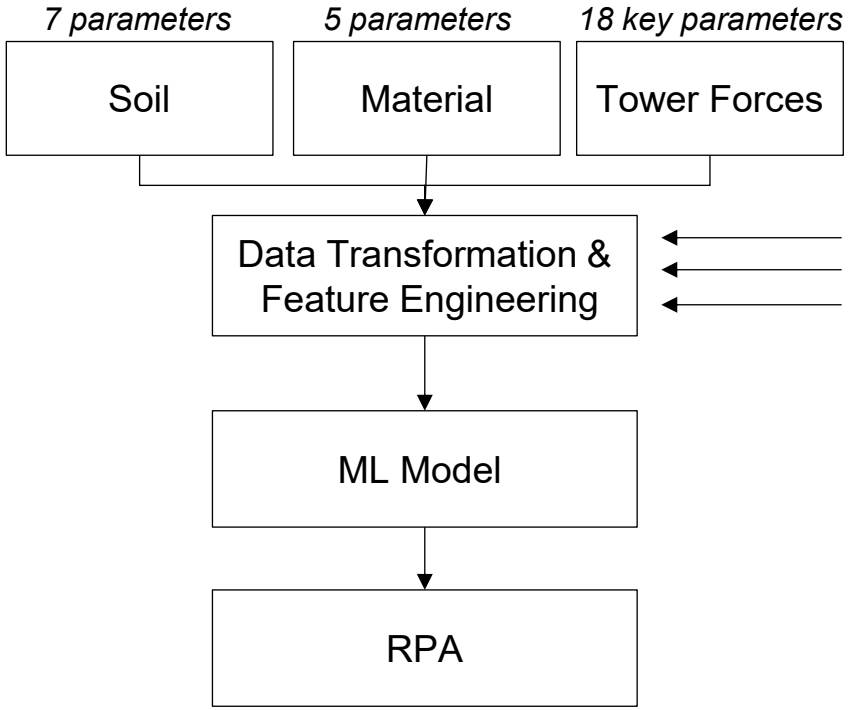


Automated processing

Leveraging Machine Learning Models

ML-Based Scientific Method

ML based Algorithm provides refined, optimal solutions compared to linear optimization methods for complex foundation construction, reducing engineering time, improving accuracy, and lowering design costs.



Total of **600+ possible design combinations** for 1 Foundation design

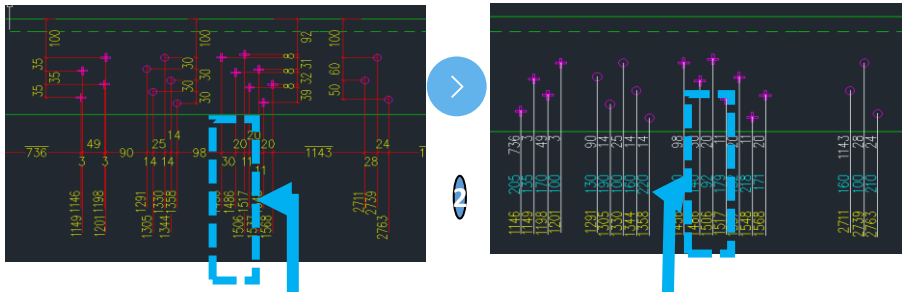
40+ Safety Design Checks

RPA Based Drawings generation

- Drawing extraction tool to eliminate repetitive manual drawing formatting operations for ease of Manufacturing & Constructability
  - **Robotic process automation** tool designed on Unified Windows Platform (UWP)
  - **Integration with AutoCAD** SDK to read and parse drawing file formats

Uncleaned Drawing (BOCAD Output) (old)

Cleaned Drawing (Auto Drawing Cleaner) (new)



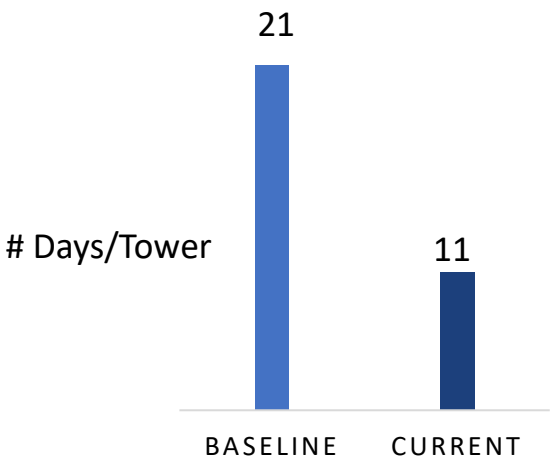
Overlaps in the shop sketches are cleaned for better readability

USE CASE / CASE STUDY

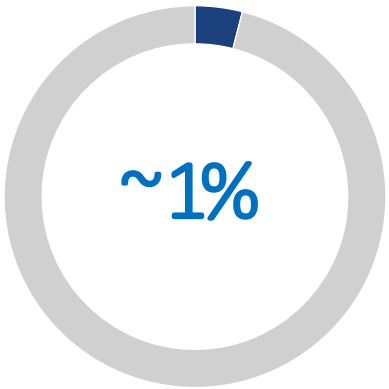
Achieved significant benefits in India based flagship project for 800 KV HVDC

- Quality of documents submitted with First-Time-Right (FTR) approval has increased to 95%

Tower structure



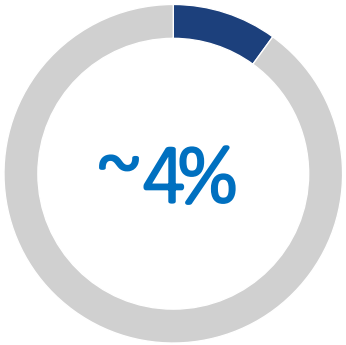
48 % Reduction in Cycle Time



Tower Weight Reduction

Foundations

50% Reduction in Cycle time



Material Cost Saving

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

www.isuw.in

Links/References (If any)